

Code Explanation

Sales Order Data Processing SQL Queries

The given SQL queries are used to process sales order data by creating intermediate and main tables. The intermediate table `sales\_orders\_comp\_stg` is created by joining data from `ord\_hdr` and `ord\_dtl` tables, while the main table `sales\_orders\_comp` is created by joining data from `sale\_orders` and other related tables.

Overview of the Code

The SQL queries create two tables: `sales\_orders\_comp\_stg` and `sales\_orders\_comp`. The first query creates an intermediate table `sales\_orders\_comp\_stg` by joining `ord\_hdr` and `ord\_dtl` tables. The second query creates the main table `sales\_orders\_comp` by joining `sale\_orders` and other related tables.

Step 1: Create Intermediate Table

This step creates an intermediate table `sales\_orders\_comp\_stg` by joining `ord\_hdr` and `ord\_dtl` tables. The join is performed on `referenceto` and `refno` columns.

```
%sql
CREATE OR REPLACE TABLE sales_orders_comp_stg AS
SELECT
    CONCAT(a.referenceto, a.refno) AS CompOrderNumber,
    ...
FROM
    (SELECT * FROM b_source.ord_hdr WHERE delete_flag <> 'D') a
LEFT JOIN
    b_source.ord_dtl b ON TRIM(a.referenceto) = TRIM(b.referenceto)
    AND TRIM(a.refno) = TRIM(b.refno)
    AND b.delete_flag <> 'D';
```

Step 2: Create Main Table

This step creates the main table `sales\_orders\_comp` by joining `sale\_orders` and other related tables. The joins are performed on various columns such as `OrderNumber`, `LineNumber`, `ShipToNumber`, etc.

```
%sql
CREATE OR REPLACE TABLE sales_orders_comp AS
SELECT
    reg.StdValue AS CompSourceSystem,
    a.OrderNumber AS CompOrderNumber,
    ...
FROM
    s_master.sale_orders a
LEFT JOIN
    s_master.sale_orders b ON a.OrderNumber = b.OrderNumber AND a.LineNumber = b.LineNumber
LEFT JOIN
    (SELECT StdValue
     FROM s_master.ord_reg
     WHERE TRIM(Type) = 'ALL'
     AND TRIM(RegCode) = 'SOURCE_SYSTEM_CODE'
     AND TRIM(sourceSystem) = 'GLOBAL')
```

```

        AND TRIM(sourceSystemValue) = 'SAP') reg ON 1=1
...
WHERE
    a.delete_flag <> 'D'
    AND YEAR(CAST(TO_DATE(FROM_UNIXTIME(UNIX_TIMESTAMP(CAST(a.process
sed_date AS STRING), 'yyyyMMdd')) AS DATE)) >= YEAR(CURRENT_DATE())
- 3;

```

Some of the key operations performed in the queries include:

- * Joining tables on various columns
- * Using `TRIM` function to remove leading and trailing spaces from column values
- * Using `CONCAT` function to concatenate column values
- * Using `CASE` statements to perform conditional logic
- * Using subqueries to retrieve data from other tables
- * Using aggregate functions such as `MAX` to retrieve maximum values

The queries also use various date and time functions such as `TO\_DATE`, `FROM\_UNIXTIME`, and `UNIX\_TIMESTAMP` to convert date and time values between different formats. The `YEAR` function is used to extract the year from a date value, and the `CURRENT\_DATE` function is used to retrieve the current date.